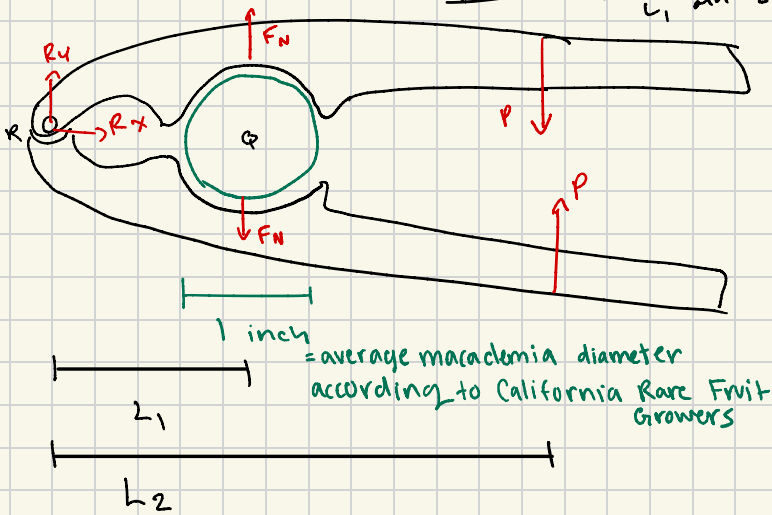
Additional problem #2

given

find: L_1 and L_2

65 lbs = max grip strength for women 20–30 according to inside tracker

(a)



= average macadamia diameter according to California Rare Fruit Growers

plan:

① moment balance about R w/ $p = 65$ lbs and $F_N = 222.18$ kg

$p = 65$ lbs in kg $\Rightarrow 29.48$ kg (upper handle)

$$\sum M_R = 222.18(L_1) - (L_2)(29.48) = 0$$

$$7.54 L_1 = L_2$$

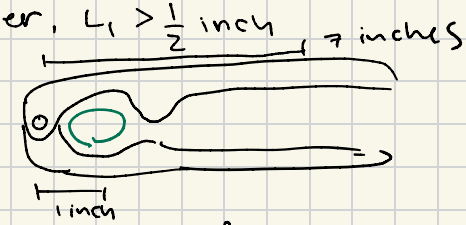
to get ratio of L_1 to L_2

w/ 1 inch avg. diameter, $L_1 > \frac{1}{2}$ inch

1 inch = L_1

~7.5 inches = L_2

$\Rightarrow$

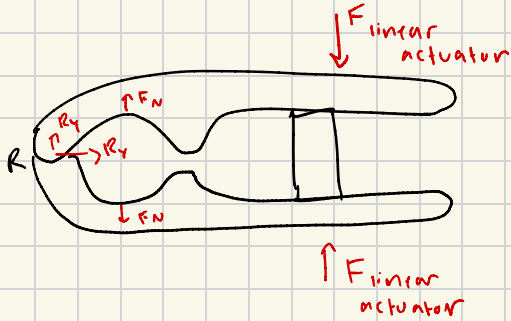


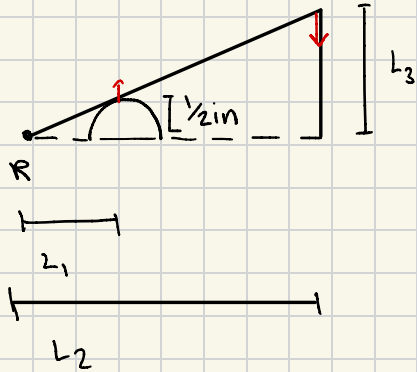
(b) The nutcracker would be usable for most men but more difficult for the average woman, especially when older.

The curret design would likely have a total length around 9 inches, we could make the design longer for better leverage + cater to more users.

9

i





$$\Sigma M_R = 0 = (L_1)(F_N) - (L_2)(F_{LA})$$

from similar triangles we know

$$\frac{L_2}{L_1} = \frac{L_3}{1/2 \text{ in}}$$

$L_3 = \frac{1}{2}$ length linear actuator when it breaks macadamia nut

or, $\frac{L_2}{L_1}$ = length

Linear actuator

IP65 Mini Linear Actuator

Model: PA-01-3-56-N-12VDC

56 lbs → 25.4012 kg

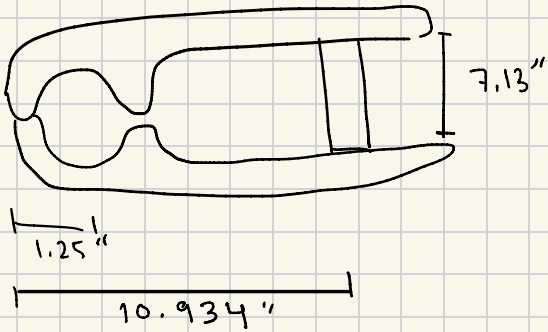
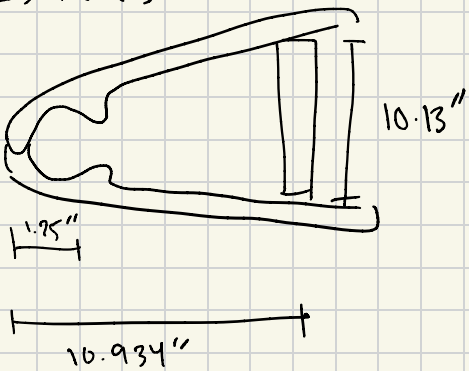
stroke = 3 inches

7.13" retracted and 10.13" extended

$\boxed{\text{does it work?}}$

$$\Sigma M_R = 0 = (L_1)(222.18) - (L_2)(25.4012)$$

$$\Rightarrow \frac{L_2}{L_1} = 8.747$$ = length of linear actuator when it breaks macadamia nut

$7.13 < 8.747 < 10.13$ ✓

Final design, making $L_1 = 1.25$ inches